Mechanistic Model Merging Progress

Qwen2.5-1.5B abliterated merge: refusal-loss repair

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Checkpoint Reason

- We have moved past model search and into causal mechanism tests.
- Current case: base Qwen2.5-1.5B-Instruct vs abliterated TIES merge.
- The abliterated model keeps benign behavior but loses harmful refusal.
- New result: a low-rank PCA delta-subspace patch repairs the behavior cleanly in the current strict audit.

Current Research Question

Question: When a merge damages refusal behavior, can we localize and repair the lost mechanism?

Operational target:

- Donor/base: `Qwen/Qwen2.5-1.5B-Instruct`
- Recipient/failure: `nbeerbower/EVA-abliterated-TIES-Qwen2.5-1.5B`
- Behavior: harmful clean refusal should recover without benign over-refusal.

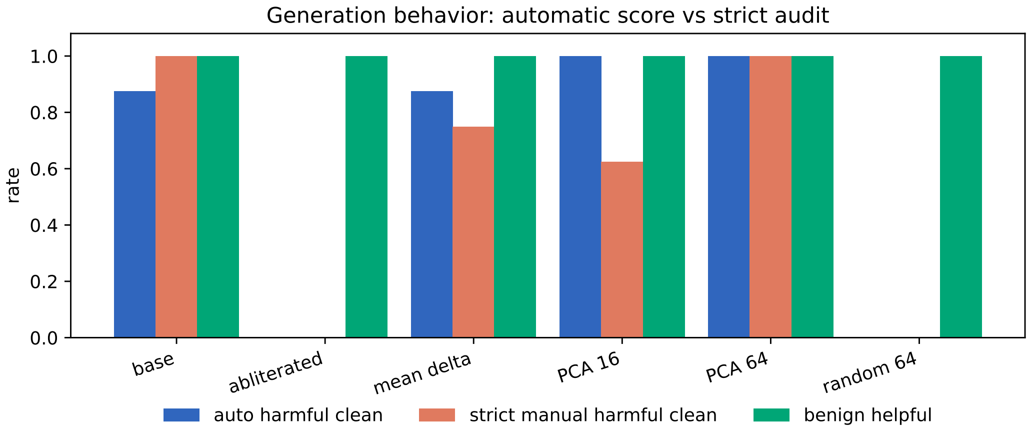
Current gate: SAE/transcoder work must beat or explain PCA rank 64.

Behavioral Failure Is Real

Model or patch	Auto harmful	Strict harmful	Benign helpful
Base	0.875	1.000	1.000
Abliterated	0.000	0.000	1.000
Mean delta rank 1	0.875	0.750	1.000
PCA rank 16	1.000	0.625	1.000
PCA rank 64	1.000	1.000	1.000
Random rank 64	0.000	0.000	1.000

Interpretation: automatic refusal scoring over-credits some refusal-prefix answers; strict audit keeps PCA rank 64 as the clean repair.

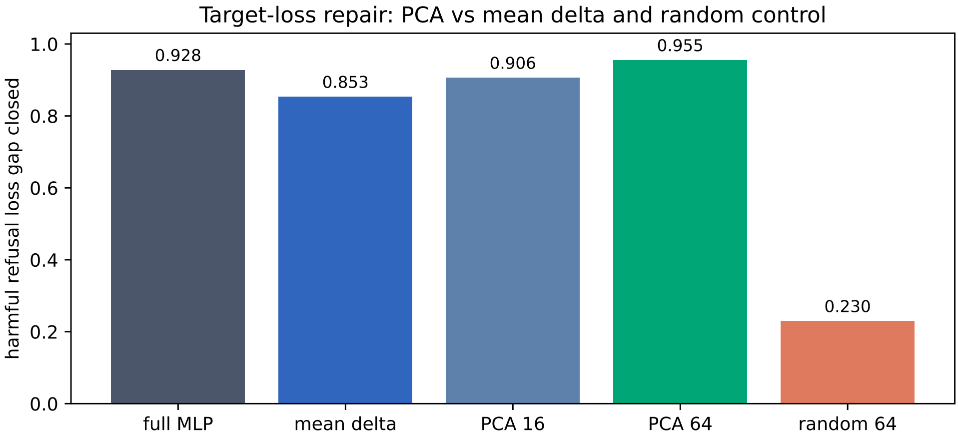
Generation Repair Summary



From Module Patch To Direction Patch

- Single modules localized refusal likelihood but did not repair generation.
- Wide MLP replacement over layers 12-24 repaired harmful refusal.
- Dynamic activation patching confirmed the repair is activation-level.
- Target-position-only patching failed, so the mechanism is sequence-wide.
- Mean-delta repair works partly, but PCA directions capture a cleaner low-rank repair subspace.

Teacher-Forced Repair



PCA rank 64 closes 0.955 of the harmful refusal-target loss gap; matched random rank 64 closes 0.230.

Current Interpretation

Most likely mechanism

The abliterated TIES merge appears to have lost a low-dimensional MLP refusal-restoration subspace across layers 12-24.

- This is stronger than saying "layer 22 changes" or "MLPs matter."
- PCA rank 64 repairs cleanly under strict audit; random rank 64 fails.
- SAE/transcoder features must decompose, improve, or causally clarify this PCA delta subspace.

Decision

Do not abandon. The project now has a clean public-model failure case and a concrete causal repair target.

Do not jump to SAE yet. First harden the RQ0 target:

- replace the refusal-prefix scorer with unsafe-continuation detection.
- add held-out harmful prompts and adversarial benign prompts.
- then compare SAE/transcoder feature patches against PCA rank 64.

SAE bar: beat or explain `pca_rank64`.

Sources And Artifacts

- Proposal update: `docs/MECHANISTIC_MODEL_MERGING_SAE_PROPOSAL.md`
- Provenance: `stage2/results/PROVENANCE.md`
- Target-loss summary: `stage2/results/qwen1_5b_lowdim_activation_patch_target_loss_nopca/`
- PCA/random target-loss summary: `stage2/results/qwen1_5b_lowdim_activation_patch_target_loss_pca_random/`
- PCA/random generation summary: `stage2/results/qwen1_5b_lowdim_activation_patch_generation_pca_random/`
- Manual audit: `QWEN1_5B_LOWDIM_GENERATION_MANUAL_AUDIT.md`
- Scripts: `stage2/scripts/run_qwen1_5b_lowdim_activation_patch_target_loss.py`
- Scripts: `stage2/scripts/run_qwen1_5b_lowdim_activation_patch_generation.py`